

3 Linear Models

3.1 Introductory remarks

- Hypothesis class of *affine* mappings in $\mathbb{R}^d$

(3.1)

- $\mathcal{H}_{d,L}$ is a set of linear functions $h_{\mathbf{w},b}(\mathbf{x})$ mapping feature vectors $\mathbf{x}$ to the scalar output variable y
- The functions $h_{\mathbf{w},b}(\mathbf{x})$ are parametrised by weight vector $\mathbf{w}$ and bias b (threshold value)
- Simpler notation by incorporating a further dimension in $\mathbf{w}' = (b, w_1, w_2, \dots, w_d)^T, \mathbf{x}' = (1, x_1, x_2, \dots, x_d)^T$
- Hypothesis class of *linear* mappings in $\mathbb{R}^{d+1}$

(3.2)

3.2 Linear Discriminant Analysis (LDA)

- Hypothesis class of half spaces for binary classification tasks with ⁽¹⁾

(3.3)

- $\mathcal{H}_{d,HS}$ is a set of generalised linear functions $\text{sgn}(h_{\mathbf{w},b}(\mathbf{x}))$ mapping feature vectors $\mathbf{x}$ to the scalar y
- The output values y should be equal to the target values t_n for the given sample

(3.4)

such that the empirical risk $L_S(h)$ (2.2) is minimal

- This can be solved, e.g. by linear programming, or perceptron algorithm, or least squares (see below)
- **Linear optimisation** (LP, linear programming)
 - In general: LP means that a linear function (objective function) $f(\mathbf{x}) = \mathbf{c} \cdot \mathbf{x}$ is minimised subject to a set of linear inequalities (restrictions) $\mathbf{A}\mathbf{x} \leq \mathbf{b}$ under admissibility conditions $\mathbf{x} \geq \mathbf{0}$
 - Sought: correct classification for each sample element

(3.5)

(Remark: the scalar product is commutative, i.e. $\mathbf{w}' \cdot \mathbf{x}' = \mathbf{x}' \cdot \mathbf{w}'$)

- An equivalent expression of the equations (3.5) are the following *homogeneous inequalities*

(3.6)

- To get *inhomogeneous inequalities*, the sample element with the smallest value of (3.6) is searched:

(3.7)

- With (3.7) the restrictions (3.6) can be formulated as a *set of inhomogeneous equations*:

(3.8)

- Easy transform: to fulfil the admissibility conditions, the data set and the restrictions must be shifted (translated) to the 1st quadrant, which is not shown here
- To find the optimal parameters $\mathbf{w}'$, a **LP solver** should be used, e.g. **simplex method** (Dantzig, 1947), or the **projective algorithm** (Karmarkar, 1984)
- **Perceptron algorithm** (Rosenblatt, 1958)
 - Iterative ERM algorithm: sequence of $\mathbf{w}(1), \mathbf{w}(2), \dots, \mathbf{w}(i), \dots$ with iteration index i
 - Simple algorithm, guaranteed to converge
 - The convergence rate is low for some data sets. Then, LP solvers are better.
 - Only if a mismatch between hypothesis output and target value happens:

(3.9)

- Then the following update of $\mathbf{w}$ is performed:

(3.10)

- This leads to an update of (3.6):

(3.11)

- Since $\|\mathbf{x}'_n\| > 0$, this leads to an increase of the left-hand side and thus tends to fulfil (3.11)

Theorem 3.1: Finite iterations within the perceptron algorithm

- .
- .
- .
- .

- The value of $\|\mathbf{w}\|_{\min}$ is important to estimate the convergence rate
- In most cases in practise, the perceptron algorithm converges fast

Theorem 3.2: VC dimension of halfspaces

The VC dimension of halfspaces in $\mathbb{R}^d$ is d , and of nonhomogeneous halfspaces in $\mathbb{R}^d$ is $d + 1$.

3.3 Multiclass LDA

- Given: set of multiple class labels
(1)

- Step function:
(2)

- Solution 1: one-versus-all

(also known as 'one-versus-rest')

- separate one class against all others
⇒ need of $(K-1)$ linear functions

(3)

- Result: the scalar output variable y
leads to **conflicts**

Alternative: binary-valued output
vectors $\mathbf{y}$

(4)

- Result: $\mathbf{y}(\mathbf{x})$ leads to more class re-
gions (2^K) than needed

- Solution 2: all combinations

- Also known as 'all pairs'
- Pairs for $K = 4$
1-2, 1-3, 1-4, 2-3, 2-4, 3-4
- Integer-valued output vector

- Majority vote: $\mathbf{y}(\mathbf{x}) \mapsto y$

⇒ $\frac{1}{2} K (K-1)$ linear separation functions

- Result: lead to conflicts, but less

- Solution 3: Discriminant values

- Real-valued output vector $\mathbf{y} \in \mathbb{R}^K$
- Criterion: maximum component

(5)

⇒ **no conflicts**

⇒ always unambiguous assignment

- Class regions are convex sets
(i.e. every feature vector
of a class region
is represented as a
convex linear combination of
two feature vectors of that region)

- Example for $K = 3$:
see page 8 of this chapter

□ Example: $K = 3$

□ Example: $K = 4$

□ Example: $K = 3$

□ Example: $K = 2$

□ Example: $K = 3$

3.4 Least Squares Linear Discriminant Analysis (LS-LDA)

- Utilisation of solution 3: real-valued output vectors $\mathbf{y}(\mathbf{x}_n) \in \mathbb{R}^K$
- Target vectors are binary-valued (1-of-K code): $\mathbf{t}_n = (0, \dots, 0, 1, 0, \dots, 0)^T$
- Objective function to minimise is the **sum of squared errors (SSE)**
 - This leads to a solution which can be represented in closed form (see below)
 - $\mathbb{E}(\mathbf{t}|\mathbf{x})$ can be well approximated
- Output vector

(3.12)

- Output and target matrix

(3.13)

- SSE using (3.13)

(3.14)

- Solution: SSE minimisation
 - First derivative with respect to $\mathbf{W}'$
 - Set this derivative equal to zero
 - Dissolve to $\mathbf{W}'$
 - These three steps are performed by calculation of the **Moore Penrose inverse** of $\mathbf{X}'$

(3.15)

- Interesting property
 - If every 1-from-K coded target vector fulfils $\mathbf{a} \cdot \mathbf{t}_n + b = 0$
 - Then the linear model also fulfils this condition for any $\mathbf{x}_n$: $\mathbf{a} \cdot \mathbf{y}(\mathbf{x}_n) + b = 0$
 - And all **components of $\mathbf{y}(\mathbf{x}_n)$** have the sum of one
 - This means that they are interpretable as **probabilities**
- Disadvantages
 - Less robust against outliers (general disadvantage of mean value approaches)
 - Weak results with small, very compact regions in the feature space
 - For this, probabilistic models are more suitable

- Side remarks (basics of linear algebra)
 - Inverse $\mathbf{A}^{-1}$:

- Generalised inverse $\mathbf{A}^-$:

- Moore Penrose inverse $\mathbf{A}^+$:

3.5 Fisher LDA

- Aim: simple dimensionality reduction, mapping onto one dimension which means a considerable loss of information
- **Given:** data set according to (3.4) **Sought:** optimal projection vector $\mathbf{w}$
- Target values for the case of two classes: $t_n \in \mathcal{T} = \{0,1\}$
and for the case of multiple classes: $t_n \in \mathcal{T} = \{0,1, \dots, K-1\}$
- Determination of index sets:

$$I_k = \{n | t_n = k\}, \quad k \in \{0,1, \dots, K-1\} \quad (3.16)$$

(1) One-dimensional separability

- **FDR, Fisher discriminant ratio** (Fisher, 1936)
 - Under the condition of equal a priori probabilities: $P_0 = P_1$, i.e. $P(t=0) = P(t=1)$
 - FDR J is proportional to the difference of the mean values
 - FDR J is inversely proportional to the mean variance

$$J = \frac{(m_1 - m_0)^2}{s_0^2 + s_1^2}, m_k = \frac{1}{|I_k|} \sum_{n \in I_k} x_n, s_k^2 = \frac{1}{|I_k| - 1} \sum_{n \in I_k} (x_n - m_k)^2 \quad (3.17)$$

(2) Multi-dimensional separability

- Projection of d -dimensional data $\mathbf{x}_n$ onto one dimension along the vector $\mathbf{w}$

$$\mathbf{x}_w = (\mathbf{w}^0 \cdot \mathbf{x}) \mathbf{w}^0, \quad \mathbf{w}^0 = \frac{1}{\|\mathbf{w}\|} \mathbf{w} \quad (3.18)$$

- These projections lead to the following projected mean vectors and projected variances

$$\text{mean}(\mathbf{w} \cdot \mathbf{x}) = \mathbf{w} \cdot \mathbf{m}_k, \quad \text{var}(\mathbf{w} \cdot \mathbf{x}) = \mathbf{w}^T \mathbf{C}_k \mathbf{w} \quad (3.19)$$

- Where the mean vectors $\mathbf{m}_k$ and the covariance matrices $\mathbf{C}_k$ of the non-projected data has been used

$$\mathbf{m}_k = \frac{1}{|I_k|} \sum_{n \in I_k} \mathbf{x}_n, \quad \mathbf{C}_k = \sum_{n \in I_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^T \quad (3.20)$$

- For the numerator of the FDR J (3.17) the following holds due to the projection

$$(m_0 - m_1)^2 = \mathbf{w}^T (\mathbf{m}_0 - \mathbf{m}_1) (\mathbf{m}_0 - \mathbf{m}_1)^T \mathbf{w} \propto \mathbf{w}^T \mathbf{C}_b \mathbf{w} \quad (3.21)$$

- For the denominator of the FDR J (3.17) the following holds due to the projection

$$s_0^2 + s_1^2 = \mathbf{w}^T \mathbf{C}_0 \mathbf{w} + \mathbf{w}^T \mathbf{C}_1 \mathbf{w} = \mathbf{w}^T (\mathbf{C}_0 + \mathbf{C}_1) \mathbf{w} = \mathbf{w}^T \mathbf{C}_w \mathbf{w} \quad (3.22)$$

- Utilising the *between classes covariance matrix* $\mathbf{C}_b$ and the *within classes covariance matrix* $\mathbf{C}_w$:
 - $\mathbf{C}_b$ expresses scattering of mean vectors of the classes around the global mean vector $\mathbf{m}$
 - $\mathbf{C}_w$ expresses scattering of data vectors of each class around the class mean vector $\mathbf{m}_k$

$$\mathbf{C}_b = \sum_{k=0}^{K-1} P_k (\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})^T, \quad \mathbf{C}_w = \sum_{k=0}^{K-1} P_k \mathbf{C}_k \quad (3.23)$$

- The FDR J for dimensional separability follows from (3.21) and (3.22)

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{C}_b \mathbf{w}}{\mathbf{w}^T \mathbf{C}_w \mathbf{w}} \quad (3.24)$$

- Maximisation of FDR: differentiate $J(\mathbf{w})$ with respect to $\mathbf{w}$ and set to zero, then: solve explicitly to $\mathbf{w}$

$$\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} = \frac{(\mathbf{w}^T \mathbf{C}_w \mathbf{w}) \mathbf{C}_b \mathbf{w} - (\mathbf{w}^T \mathbf{C}_b \mathbf{w}) \mathbf{C}_w \mathbf{w}}{|\mathbf{w}^T \mathbf{C}_w \mathbf{w}|^2} = 0$$

With $c_1 = \mathbf{w}^T \mathbf{C}_b \mathbf{w}$, $c_2 = \mathbf{w}^T \mathbf{C}_w \mathbf{w}$ it follows: $c_1 \mathbf{C}_w \mathbf{w} = c_2 \mathbf{C}_b \mathbf{w} = c_2 c_3 (\mathbf{m}_0 - \mathbf{m}_1)$

Where the following projection holds: $\mathbf{C}_b \mathbf{w} = c_3 (\mathbf{m}_0 - \mathbf{m}_1)$

- Solution

$$\mathbf{w} = \frac{c_2 c_3}{c_1} \mathbf{C}_w^{-1} (\mathbf{m}_0 - \mathbf{m}_1) \quad (3.22)$$

3.6 Multiple Linear Regression

- Given: Sample of random variables $\mathbf{x}_n \in \mathcal{X} \subset \mathbb{R}^d$ and random target vectors $\mathbf{t}_n \in \mathcal{T} \subset \mathbb{R}^l$
- Sought: functional relation, mapping $h: \mathbf{x} \mapsto \mathbf{t}$, here: parameters of a linear mapping h
- Approach
 - Association task: target vectors $\mathbf{t}$; requests regression task to be executed several times in succession
 - Regression task: one random variable $t \in \mathcal{T} \subset \mathbb{R}^1$ is considered to be a dependent variable
 - Unknown (random) influences are described with an additive term ε (expectational error)
 - Linear/Nonlinear regression: the function h is linear/nonlinear and explains the response (target) t except for one residual error: ⁽¹⁾

Linear regression model

- Regressor $\mathbf{x}$ is mapped linearly to the target values t

(3.26)

- Where $\mathbf{w}$ is again a weight vector and ε is often referred to as a stochastic environment
- $\mathbf{x}, \mathbf{w}, t, \varepsilon$ are realisations of the random variables X, W, T, E
- Given: N realisations (observations) of X, T
- Sought: $\mathbf{w}$ as a realisation of W
- Assumption: $\mathbf{x}, t$ have zero mean

MAP estimate of the weight vectors (maximum a posteriori)

- Regressor X acts as an excitation and has no relationship to W
- Information about W is contained exclusively in T
 - joint probability density function (jpdf) between W and T given X is searched for:

(3.27)

- **Bayes theorem** (Thomas Bayes, 1736)

(3.28)

- Where the following pdf's are used:
 - Observation pdf: $p(t|\mathbf{w}, \mathbf{x})$
 - A posteriori pdf: $p(\mathbf{w}|\mathbf{x}, t)$
 - A priori pdf: $p(\mathbf{w})$
 - Evidence pdf: $p(t)$
- Oftentimes **likelihood functions** $l(\mathbf{w}|\mathbf{x}, t)$ are used to find the solution
 - Less rigorous concept than pdf: $p(\mathbf{w}|\mathbf{x}, t) \sim l(\mathbf{w}|\mathbf{x}, t)p(\mathbf{w})$
 - Maximum likelihood estimation: $\mathbf{w}_{ML} = \operatorname{argmax} l(\mathbf{w}|\mathbf{x}, t)$ (3.30)
 - Maximum a posteriori estimation: $\mathbf{w}_{MAP} = \operatorname{argmax} p(\mathbf{w}|\mathbf{x}, t)$ (3.31)
 - MAP is more substantiated than ML estimation
 - MAP contains $p(\mathbf{w})$ and thus all imaginable information about $\mathbf{w}$
 - ML ignores any a priori information
 - ML is based only on observation model $(\mathbf{x}, t) \rightarrow$ non-unique solutions
 - Improvements are possible by considering a priori information $p(\mathbf{w}) \rightarrow$ MAP
- Logarithmic ML often lead to mathematical simplifications:
 - 1) Likelihood functions often come from exponential family: $l(\mathbf{x}) \sim e^{ax} \rightarrow L(\mathbf{x}) = \ln l(\mathbf{x}) \sim x$
 - 2) Products of pdf become sums of pdf
- Log likelihood estimate of (3.28):

$$\mathbf{w}_{MAP} = \operatorname{argmax} \ln(p(\mathbf{w}|\mathbf{x}, t)) \quad (3.31)$$

Parameter estimation

- Given: sample $S = \{(\mathbf{x}_n, t_n) | \mathbf{x}_n \in \mathbb{R}^d, t_n \in \mathbb{R}, n \in [N]\}$
- Assumptions
 - 1) Stochastic environment ε is under normal distribution (Gaussian pdf)
 - 2) Sample S is drawn i.i.d. (independent and identically distributed): $S \sim \mathcal{D}^N$
 - 3) Residuals ε_n are also normally distributed: $\varepsilon_n \sim N(0, \sigma_\varepsilon)$, $p(\varepsilon_n) = \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \exp\left(-\frac{\varepsilon_n^2}{2\sigma_\varepsilon^2}\right)$, $n = 1, \dots, N$
 - 4) Stationarity: all parameters $\mathbf{w}$ are constant in time, every component of $\mathbf{w}$ has Gaussian a priori pdf:

$$w_k \sim N(0, \sigma_w^2), p(w_k) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left(-\frac{w_k^2}{2\sigma_w^2}\right) \quad \forall k = 1, 2, \dots, d$$

- Linear model (3.26) applied to the n^{th} observation: $t_n = \mathbf{w} \cdot \mathbf{x}_n + \varepsilon$, $n \in [N]$
 - 1st moment: $\mathbb{E}(T) = \mathbf{w} \cdot \hat{\mathbf{x}}$
 - 2nd moment: $\text{var}(T) = \mathbb{E}(T - \mathbb{E}(T))^2 = \mathbb{E}(\varepsilon^2) = \sigma_\varepsilon^2$
- This results in the following **likelihood function**:

$$l(\mathbf{w}|\mathbf{x}_n, t_n) = \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \exp\left(-\frac{1}{2\sigma_\varepsilon^2} (t_n - \mathbf{w} \cdot \mathbf{x}_n)^2\right) \quad \forall n \in [N] \quad (3.32)$$

- Due to assumption 2) the following **likelihood function of the whole experiment** results:

$$\begin{aligned} l(\mathbf{w}|\mathbf{x}, t) &= \prod_{n=1}^N l(\mathbf{w}|\mathbf{x}_n, t_n) = \frac{1}{(\sqrt{2\pi\sigma_\varepsilon^2})^N} \prod_{n=1}^N \exp\left(-\frac{1}{2\sigma_\varepsilon^2} (t_n - \mathbf{w} \cdot \mathbf{x}_n)^2\right) \\ &= \frac{1}{(\sqrt{2\pi\sigma_\varepsilon^2})^N} \exp\left(-\frac{1}{2\sigma_\varepsilon^2} \sum_{n=1}^N (t_n - \mathbf{w} \cdot \mathbf{x}_n)^2\right) \end{aligned} \quad (3.33)$$

- This means that $\mathbf{w}$ represents knowledge of the complete sample S
- Due to assumption 4) the following **a priori pdf** results:

$$p(\mathbf{w}) = \prod_{k=1}^N p(w_k) = \frac{1}{(\sqrt{2\pi\sigma_w^2})^N} \prod_{k=1}^N \exp\left(-\frac{w_k^2}{2\sigma_w^2}\right) = \frac{1}{(\sqrt{2\pi\sigma_w^2})^N} \exp\left(-\frac{1}{2\sigma_w^2} \sum_{k=1}^N w_k^2\right) \quad (3.34)$$

- Due to the Bayes theorem the following **a posteriori pdf** results:

$$p(\mathbf{w}|\mathbf{x}, t) \propto \exp\left(-\frac{1}{2\sigma_\varepsilon^2} \sum_{n=1}^N (t_n - \mathbf{w} \cdot \mathbf{x}_n)^2 - \frac{\|\mathbf{w}\|^2}{2\sigma_w^2}\right) \quad (3.35)$$

- After a longer derivation: MAP estimation (3.30) of a-posteriori pdf (3.35) gives the following **solution**:

$$(3.36)$$

- Where $\hat{\mathbf{R}}_{XX}$ is the time averaged $d \times d$ auto-correlation matrix of the regressor X :

$$(3.37)$$

- Where λ is the following regularisation parameter: $\lambda = \sigma_\varepsilon^2 / \sigma_w^2$
- And where $\hat{\mathbf{r}}_{XT}$ is the time averaged $d \times 1$ cross correlation function of X with T :

$$(3.38)$$

- Due to assumption 1) all a-priori pdf of each component in $\mathbf{w}$ are identical: $p(w_k) = p_w$
 $\Rightarrow$ solution (3.36) simplifies to:

$$\mathbf{w}_{ML} = \hat{\mathbf{R}}_{XX}^{-1} \hat{\mathbf{r}}_{XT} \quad (3.36)$$

- This means that the ML estimate (3.39) is based solely on the observation model
 - Under the assumption that the sample elements $\{(\mathbf{x}_n, t_n)\}$ are outcomes of a joint ergodic process, the ML estimate (3.39) is asymptotically unbiased: $\lim_{N \rightarrow \infty} \hat{\mathbf{w}}_{ML} = \mathbf{w}$
- The MAP estimate (3.36) is biased
 - Due to the inclusion of pre-knowledge a regularisation of the ML estimate is introduced
 - Bias stability trade-off: one has to choose between high stability, high bias and low stability, low bias
 - Using the regularisation parameter as control
 - $\lambda = 0 \rightarrow$ full confidence in the observation model, $\lambda \rightarrow \infty \rightarrow$ no confidence in the observation model
- Other name for MAP estimate: regularised least squares (RLS)

Multi-Class LDA

Example of **solution 3** in $\mathbb{R}^2$

$K = 3$

